The sharp Berry–Esseen bound for independent arrays with small Lyapunov ratio

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Abstract

We prove that every finite family of independent centered real random variables $X_1, \dots, X_n$ with positive total variance $V = \sum_{i=1}^n \mathbb{E}X_i^2$ and finite third absolute moments $R = \sum_{i=1}^n \mathbb{E}|X_i|^3$ satisfies the sharp Berry–Esseen inequality $\Delta \le C_{\mathrm{E}} L$ whenever the Lyapunov ratio $L = R/V^{3/2}$ satisfies $L \le \exp(-\exp(97))$, where $\Delta = \sup_x |\mathbb{P}(V^{-1/2}\sum_{i=1}^n X_i \le x) - \Phi(x)|$ and $C_{\mathrm{E}} = (3+\sqrt{10})/(6\sqrt{2\pi})$ is Esseen's constant. The constant C_{E} is optimal. Our proof adapts the variational, uniform-jitter, and local-comparison framework introduced by He and Cheng for identically distributed summands to independent summands with arbitrary distributions. A penalized optimization selects an attained counterexample and yields variational contact equations for each summand. A finite pigeonhole argument identifies an approximate lattice period, and a quantitative stability form of Esseen's moment inequality confines every summand either to two narrow clusters separated by that period or to a small neighborhood of zero with small total variance. A two-regime conditional comparison against a common-diameter Bernoulli array then rules out any violation of $\Delta \le C_{\mathrm{E}} L$.

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1 Introduction and main result

1.1 The theorem and its setting

Let $X_1, \dots, X_n$ be independent centered real random variables with positive total variance and finite third absolute moments. Write

$$V = B^2 = \sum_{i=1}^{n} \mathbb{E}X_i^2 > 0, \qquad R = T = \sum_{i=1}^{n} \mathbb{E}|X_i|^3 < \infty, \qquad L = \ell = \frac{T}{B^3}, \tag{1.1}$$

and define the Kolmogorov normal-approximation error and Berry–Esseen ratio by

$$\Delta = \sup_{x\in\mathbb{R}} \left| \mathbb{P}\left\{ B^{-1} \sum_{i=1}^{n} X_i \le x \right\} - \Phi(x) \right|, \qquad \mathcal{R}(X_1, \dots, X_n) = \frac{\Delta}{\ell},$$

where Φ and ϕ are the standard normal distribution function and density. Deterministic zero summands are permitted. Put $c_{\mathrm{E}} = 3 + \sqrt{10}$ and $C_{\mathrm{E}} = c_{\mathrm{E}}\phi(0)/6 = (3 + \sqrt{10})/(6\sqrt{2\pi})$.

Theorem 1.1. *For every finite independent array satisfying* (1.1)*,*

$$\boxed{\ell \le \ell_0 := \exp(-\exp(97)) \quad \Longrightarrow \quad \Delta \le C_{\mathrm{E}}\ell.}$$

The constant C_{E} is optimal: no smaller constant holds on $0 < \ell \le \ell_0$, even when all summands are identically distributed two-point variables.

Theorem 1.1 extends to non-identically distributed summands the result of He and Cheng [6, Theorem 1.1], who proved that identically distributed summands satisfy $\Delta \le C_{\mathrm{E}}\ell$ once the sample size n exceeds a universal threshold. For general independent arrays, the number of summands n can be increased arbitrarily without changing the sum or the Lyapunov ratio simply by adjoining deterministic zero summands; the intrinsic measure of proximity to the asymptotic regime is therefore the Lyapunov ratio ℓ.

1.2 Open questions and relationship to companion papers

The threshold $\ell_0 = \exp(-\exp(97))$ in Theorem 1.1 is an explicit sufficient bound resulting from tracking concrete constants through the period-selection and conditional-comparison stages of the proof. Whether the sharp inequality $\Delta \le C_{\mathrm{E}}\ell$ holds for *all* $\ell > 0$—that is, whether $C_{\mathrm{ind}} = C_{\mathrm{E}}$ holds unconditionally—remains open in the intermediate regime $\ell_0 < \ell < 1.321$.

Two quantitative results from companion papers in this collection enter the proof of Theorem 1.1:

$$\mathcal{R}(h(B_1 - p_1), \dots, h(B_m - p_m)) \le C_{\mathrm{E}}, \tag{1.2}$$

$$\mathcal{R}(X_1, \dots, X_n) < \frac{9}{20}. \tag{1.3}$$

In (1.2), $B_1, \dots, B_m$ are independent Bernoulli variables with arbitrary parameters $p_i \in (0,1)$ and $h > 0$; this is the common-diameter theorem [1, Theorem 1.2(ii)], which extends Schulz's binomial theorem [7, Theorem 1] to unequal probabilities. Inequality (1.3) follows from the unconditional upper bound $C_{\mathrm{ind}} < 0.44988 < 9/20$ [3, Theorem 1.1]. We also use Shevtsova's asymptotic remainder estimate for independent summands [8, Corollary 4.18, p. 303]:

$$\Delta \le C_{\mathrm{E}}\ell + 0.2538\ell^{4/3} \qquad (\ell \le 10^{-3}). \tag{1.4}$$

In addition, a companion structural paper [2] proves that below an ineffective Lyapunov threshold $\varepsilon_0 > 0$, every n-summand global maximizer of $\mathcal{R}$ consists of identically distributed two-point summands and zero summands. Theorem 1.1 establishes an explicit threshold ℓ_0 for the value inequality $\Delta \le C_{\mathrm{E}}\ell$ across all independent arrays.

1.3 Architecture of the proof

Our proof develops the three core mechanisms of He and Cheng [5, Sections 2–5 and Appendix A]—variational support equations via coordinate contamination, approximate lattice period detection via uniform jitter, and a two-regime local comparison—in forms that hold uniformly for non-identically distributed summands. Set

$$p_{\mathrm{E}} = \frac{4 - \sqrt{10}}{2} = \frac{1 - 1/c_{\mathrm{E}}}{2}, \qquad q_{\mathrm{E}} = 1 - p_{\mathrm{E}}, \qquad \tau_{\mathrm{E}} = p_{\mathrm{E}}^2 + q_{\mathrm{E}}^2, \qquad h_{\mathrm{E}} = \frac{1}{\tau_{\mathrm{E}}}. \tag{1.5}$$

By Esseen's moment inequality [4, pp. 160–163], any centered random variable Y supported on a coset of $h\mathbb{Z}$ satisfies

$$|\mathbb{E}Y^3| + 3h\mathbb{E}Y^2 \le c_{\mathrm{E}}\mathbb{E}|Y|^3, \tag{1.6}$$

with equality for nondegenerate Y if and only if $Y \overset{d}{=} \pm h(B - p_{\mathrm{E}})$ with $B \sim \mathrm{Bern}(p_{\mathrm{E}})$.

Writing $F_S(x) = \mathbb{P}(S \le x)$ and $F_S^-(x) = \mathbb{P}(S < x)$, continuity of Φ gives

$$\sup_{x \in \mathbb{R}} |F_S(x) - \Phi(x/B)| = \sup_{x \in \mathbb{R}} \max\{F_S(x) - \Phi(x/B),\ \Phi(x/B) - F_S^-(x)\}, \tag{1.7}$$

and reflection $X_i \mapsto -X_i$ interchanges the two one-sided branches. The proof proceeds in five stages:

(1) **Penalized selection and variational contact equations (Section 2):** If an n-summand array violated $\Delta \le C_{\mathrm{E}}\ell$ at some $\ell \le t$, maximizing $[F_S(z) - \Phi(z)]/\ell - \psi_t(\ell)$ over n-summand variance-one arrays for a smooth penalty $\psi_t(\ell)$ (chosen to dominate (1.4) and (1.3) for $\ell \ge 2t$) yields an attained maximizer with $\ell < 2t$ and ratio $R > C_{\mathrm{E}}$. Perturbing each summand X_i toward a point mass δ_y yields a variational contact inequality (2.4) with equality on $\operatorname{supp} X_i$, which implies the uniform support bound $\operatorname{supp} X_i \subset (-8\ell, 8\ell)$ (Lemma 2.3).

(2) **Approximate period selection and uniform jitter (Section 3 and Appendix A):** Applying a finite pigeonhole lemma (Lemma 3.1) to the symmetrized quadratic-variation measure of $(X_1/\ell, \dots, X_n/\ell)$ produces a scale $0 \le h \le 128\pi$ such that convolving S with an independent uniform variable $\ell U_h \sim \mathrm{Unif}[-\ell h/2, \ell h/2]$ damps all high-frequency peaks and brings $F_{S+\ell U_h}$ within $\rho\ell$ of its one-term Edgeworth expansion (Proposition 3.2).

(3) **Quantitative Esseen stability and two-cluster confinement (Section 4 and Appendix B):** Rounding each scaled summand $Y_i = X_i/\ell$ to its nearest lattice coset of span h and applying a quantitative stability refinement of Esseen's inequality (1.6) forces h close to h_{E} and the maximizing threshold z close to 0. Comparing the one-sided jitter smoothing loss across deletions restricts all pairwise differences of support points of Y_i to a γ-neighborhood of $h\mathbb{Z}$, while the cubic contact polynomial bounds $\mathrm{diam}(\mathrm{supp}\, Y_i) < 7/2$. This splits the summands into $m \ge 1/(4\ell^2)$ two-cluster variables $Y_i = h_i(B_i - p_i) + U_i$ (with $p_i \in [1/4, 3/4]$, $|h_i/h - 1| \le \lambda$, and $|U_i| \le \lambda h$) and small-diameter variables Z_j with $|Z_j| \le \lambda h$ and total variance $\sum_j \mathrm{Var}\, Z_j \le \lambda m h^2$ (Proposition 4.1).

(4) **Local comparison near a common diameter (Section 5 and Appendix C):** Theorem 5.1 shows that every array satisfying the two-cluster reduction with $m \ge N$ and $\lambda = e^{-\mathcal{A}}$ ($\mathcal{A} = 38e^{43}$) obeys $\mathcal{R} \le C_{\mathrm{E}}$. Rescaling by the variance-weighted mean diameter $\bar{h}$, the total residual variance $\Lambda = \Lambda_A + \Lambda_U$ measures both diameter spread and intra-cluster noise. When $\Lambda = 0$, the array has a common diameter and (1.2) applies. When $0 < \Lambda \le e^{-43}$, conditioning on the Bernoulli count $K_0 = \sum_{i=1}^m B_i$ and applying a one-sided clipping inequality shows that the residual W strictly reduces the ratio below that of the common-diameter reference array. When $\Lambda \ge e^{-43}$, exponential tilting of the Bernoulli labels and a conditional Fourier minorant show that $S - x$ places mass at least $q/\sqrt{m}$ in both $(1/4, 1/2)$ and $(-1/2, -1/4)$, creating a one-sided jitter smoothing loss that pushes $\mathcal{R}$ strictly below C_{E}.

(5) **Threshold evaluation (Section 6):** Composing the explicit parameter dependencies shows that $\ell_0 = \exp(-\exp(97)) < t_*$, completing the proof of Theorem 1.1.

2 Penalized selection and variational support bounds

Normalize total variance to $B^2 = 1$, so that $\ell = T = \sum_{i=1}^n \mathbb{E}|X_i|^3$. To obtain an attained counterexample from a hypothetical small-ℓ violation, we maximize the one-sided ratio minus a smooth penalty on ℓ at a fixed number of summands n.

Lemma 2.1 (Localized counterexample selection)**.** *Let $0 < \eta \le 1/100$ and $0 < t \le \min\{1/2000, \eta^3/108\}$. If a finite independent array has $\ell \le t$ and $\mathcal{R} > C_{\mathrm{E}}$, then there exists an attained maximizer of the penalized objective (2.3) with the same number of summands, a threshold $z \in \mathbb{R}$, and full ratio $R = \mathcal{R}$ satisfying*

$$C_{\mathrm{E}} < R < \frac{9}{20}, \qquad \ell < 2t, \qquad R \le K := R + \ell\psi_t'(\ell) \le R + \eta. \tag{2.1}$$

Moreover, if a summand of this maximizer has variance s and third absolute moment b, then

$$\psi_t\left(\frac{\ell - b}{(1-s)^{3/2}}\right) - \psi_t(\ell) \le 2\eta s. \tag{2.2}$$

Proof. Define $f(v) = 0$ for $v \le 0$, $f(v) = v^2/(1+v)$ for $v > 0$, and $\psi_t(u) = 2u^{1/3} f(\log(u/t))$. Then ψ_t is continuously differentiable, nonnegative, nondecreasing, and identically zero on $(0, t]$. After

reflecting the array if necessary, the supremum of

$$\frac{F_S(z)-\Phi(z)}{\ell}-\psi_t(\ell) \tag{2.3}$$

over centered variance-one arrays of length n and $z\in\mathbb{R}$ exceeds C_{E}. Along any maximizing sequence with objective value at least $r_0>C_{\mathrm{E}}$, we have $n^{-1/2}\le\ell\le r_0^{-1}$, so the third absolute moments are uniformly bounded. Coordinatewise weak compactness, uniform integrability of second moments, boundedness of z (since $F_S(z)-\Phi(z)\ge r_0/\sqrt{n}$), upper semicontinuity of $F_S(z)$ at moving thresholds, and lower semicontinuity of ℓ yield an attaining limit array with ℓ preserved (following the fixed-n compactness argument of He and Cheng [5, Lemma 4.1]).

Since $f(\log 2)>4/15>1/4$, we have $\psi_t(u)>u^{1/3}/2$ for $u\ge 2t$. For $2t\le u\le 10^{-3}$, $\psi_t(u)>0.2538u^{1/3}$, so (1.4) implies that the objective in (2.3) is at most C_{E}; for $u\ge 10^{-3}$, $\psi_t(u)>1/20>9/20-C_{\mathrm{E}}$, so (1.3) likewise bounds the objective below C_{E}. Thus the attained maximizer must satisfy $\ell<2t$. For $t<u\le 4t$, $f<1$ and $0\le f'\le 1$ give

$$0\le u\psi_t'(u)\le\frac{8}{3}(4t)^{1/3}<3(4t)^{1/3}\le\eta,$$

which also holds on $(0,t]$. Every summand satisfies $s^{3/2}\le\ell<10^{-3}$, so $s\le 1/4$ and $\ell_-=(\ell-b)/(1-s)^{3/2}<(8/5)\ell<4t$. When $\ell_->\ell$, integrating $u\psi_t'(u)\le\eta$ over $[\ell,\ell_-]$ gives

$$\psi_t(\ell_-)-\psi_t(\ell)\le\eta\log(\ell_-/\ell)\le\frac{3}{2}\eta[-\log(1-s)]\le 2\eta s,$$

proving (2.2). □

For each summand X_i of the selected maximizer, write $s_i=\mathbb{E}X_i^2$, $b_i=\mathbb{E}|X_i|^3$, $M_i=\mathbb{E}[X_i|X_i|]$, and let G_i be the distribution function of the deleted sum $S-X_i$.

Lemma 2.2 (Variational support contact)**.** *For every $i\in\{1,\dots,n\}$ and $y\in\mathbb{R}$,*

$$\begin{aligned}D_i(y):=G_i(z-y)-F_S(z)+\phi(z)y+\frac{z\phi(z)}{2}(y^2-s_i)\\ +\frac{3}{2}K\ell(y^2-s_i)-K\{|y|^3-b_i-3M_iy\}\le 0,\end{aligned} \tag{2.4}$$

with equality $D_i(y)=0$ at every $y\in\operatorname{supp}X_i$.

Proof. Replacing $\mathrm{Law}(X_i)$ by $(1-e)\,\mathrm{Law}(X_i)+e\delta_y$ $(0\le e<1)$, keeping the uncentered threshold z fixed, and recentering and restandardizing the sum gives right derivatives y, y^2-s_i, and $|y|^3-b_i-3M_iy$ for the mean, total variance, and third absolute moment sum at $e=0$, while the uncentered CDF has right derivative $G_i(z-y)-F_S(z)$. Differentiating (2.3) at $e=0$ and multiplying by ℓ gives $D_i(y)\le 0$ by maximality. Since $\mathbb{E}D_i(X_i)=0$ and D_i is upper semicontinuous, $D_i(y)=0$ holds for every $y\in\operatorname{supp}X_i$ (compare He and Cheng [5, Lemma 4.3]). □

Lemma 2.3 (Uniform support and moment bounds)**.** *Every nonzero summand X_i of the selected maximizer satisfies*

$$b_i<3\ell s_i,\qquad s_i<9\ell^2,\qquad \operatorname{supp}X_i\subset(-8\ell,8\ell). \tag{2.5}$$

Proof. Replacing X_i by 0 leaves an admissible n-summand array whose penalized objective is at most $R-\psi_t(\ell)$, so its unpenalized ratio at every threshold is at most $R+2\eta s_i$ by (2.2). Comparing

the deleted normal approximation convolved with X_i against its convolution with $N(0, s_i)$ via a second-order Taylor expansion gives, with $a = \phi(0)(1 + 4\phi(0))/6 < 13/75$,

$$R\ell(1 - s_i)^{3/2} \le (R + 2\eta s_i)(\ell - b_i) + ab_i.$$

Using $1-(1-s_i)^{3/2} \le 3s_i/2$ and rearranging yields $(R-a)b_i \le (3R/2+2\eta)\ell s_i < 3(R-a)\ell s_i$ (since $R > 2/5$ and $\eta \le 1/100$), proving $b_i < 3\ell s_i$ and $s_i^{3/2} \le b_i < 3\ell s_i$, so $s_i < 9\ell^2$.

For $0 \le s \le 1/4$, define the Gaussian scale-translation remainder

$$H_s(z, y) = \Phi\Big(\frac{z - y}{\sqrt{1 - s}}\Big) - \Phi(z) + \phi(z)y + \frac{1}{2}z\phi(z)(y^2 - s),$$

which satisfies the global and local Taylor bounds

$$|H_s(z, y)| \le \frac{\phi(0)}{6}|y|^3 + \frac{\phi(0)}{2}s|y| + \frac{s^2}{6}, \tag{2.6}$$

$$\left|H_s(z, y) - \phi''(z)\left(\frac{sy}{2} - \frac{y^3}{6}\right)\right| \le \frac{y^4}{36} + \frac{sy^2}{6} + \frac{s^2}{6}. \tag{2.7}$$

Bounding $G_i(z - y)$ in the contact equality $D_i(y) = 0$ by the deleted normal approximation and using $(1 - s)^{-3/2} < 8/5$ and $(1 - s)^{-3/2} - 1 - 3s/2 \le 6s^2$ shows that every $y \in \operatorname{supp} X_i$ obeys

$$0 \le H_s(z, y) - K|y|^3 + \frac{3}{2}K\ell y^2 + 3KM_i y + 6R\ell s^2 + \eta b + 4\eta\ell s. \tag{2.8}$$

Let $A_i = \max_{y\in\operatorname{supp} X_i} |y|$. Using $|M_i| \le s_i$, $b_i \le s_i A_i$, $K \le 23/50$, $s_i \le 9\ell^2$, $\ell \le 1/8$, and (2.6) in (2.8) at $|y| = A_i = x\ell$ gives

$$\frac{x^3}{3} \le \frac{7}{10}x^2 + \frac{72}{5}x + \frac{1287}{200},$$

which implies $x < 8$, and hence $A_i < 8\ell$. □

3 Approximate period selection and uniform jitter

Convolving S with an independent uniform random variable suppresses the high-frequency peaks of $\mathbb{E}e^{\mathrm{i}tS}$ provided the uniform jitter width is matched to any approximate lattice period shared by the summands. The following finite pigeonhole lemma selects such a period directly from a probability measure μ on $[-M, M]$.

Lemma 3.1 (Finite approximate-period selection)**.** *Let μ be a probability measure on $[-M, M]$ ($M > 0$), and define*

$$g(u) = \int_{[-M,M]} \frac{|e^{\mathrm{i}ud} - 1|^2}{d^2}\,\mu(dd), \qquad q(u) = \sqrt{g(u)}, \qquad \delta = \frac{1}{4M},$$

where the integrand at $d = 0$ is u^2. For $T \ge \delta$ and $0 < r \le \delta/4$, put

$$B_0 = \frac{T}{\delta} + 2, \qquad J = \lceil \log_2(T/\delta) \rceil + 1, \qquad \varepsilon = \frac{r}{2B_0^J}, \qquad E = \{u \in [\delta, T] : q(u) \le \varepsilon\}.$$

If $E \ne \varnothing$, there exists $p \in [\delta, T]$ such that $q(p) \le r/2$ and $\operatorname{dist}(u, p\mathbb{Z}) < r$ for every $u \in E$. Setting $h = 2\pi/p$, we have for all $\ell > 0$,

$$\int_\delta^T e^{-g(u)/(2\ell^2)} |\operatorname{sinc}(hu/2)|\,\frac{du}{u} \le \frac{56\sqrt{\pi}}{9}\frac{r\ell}{p^2} + \log(T/\delta)e^{-\varepsilon^2/(2\ell^2)}. \tag{3.1}$$

If $E = \varnothing$, (3.1) holds with $h = 0$ and only the second term on the right-hand side.

Proof. By the triangle inequality in $L^2(\mu)$, q is even, subadditive, and 1-Lipschitz; moreover $g''(u) \geq 2 - M^2 g(u)$ and $g(u) \geq u^2/2$ on $[0, \delta]$. If $E \neq \varnothing$, pick $p_0 \in E$. Whenever there exists $v \in E$ with $\operatorname{dist}(v, p_j\mathbb{Z}) \geq r$, let k be the nearest integer to v/p_j and set $p_{j+1} = |v - kp_j|$. As long as $p_j \geq \delta$, we have

$$r \leq p_{j+1} \leq \frac{p_j}{2}, \qquad q(p_{j+1}) \leq \varepsilon + \left(\frac{T}{\delta} + \frac{1}{2}\right) q(p_j) \leq B_0^{j+1}\varepsilon.$$

Because any point in $[r, \delta]$ satisfies $q(u) \geq u/\sqrt{2} \geq r/\sqrt{2} > r/2 \geq B_0^J \varepsilon$, at most $J - 1 = \lceil \log_2(T/\delta) \rceil$ halving steps can occur before the process terminates at a period $p \in [\delta, T]$ with $q(p) \leq r/2$ and $\operatorname{dist}(u, p\mathbb{Z}) < r$ for all $u \in E$.

For each $k \geq 1$ such that $I_k = [kp - r, kp + r] \cap [\delta, T]$ intersects E, we have $q \leq \varepsilon + 2r \leq 5\delta/8$ on I_k, so $g'' \geq 1$ and $g(u) \geq (u - a_k)^2/2$ around the minimum a_k of g on I_k. Since $|\operatorname{sinc}(\pi u/p)| \leq r/u$ on I_k and $u \geq (k - 1/4)p$, integrating over I_k and summing $\sum_{k \geq 1} (k - 1/4)^{-2} \leq 28/9$ yields the first term in (3.1), while $g(u) > \varepsilon^2$ outside $\bigcup_k I_k$ yields the second. □

Proposition 3.2 (Uniform jitter Edgeworth approximation)**.** *Suppose* $\sum_{i=1}^n \mathbb{E}X_i^2 = 1$, $\ell = \sum_{i=1}^n \mathbb{E}|X_i|^3 \leq 1/8$, *and* $|X_i| \leq 8\ell$ *for all* i. *For* $0 < \rho \leq 1$, *define*

$$\begin{aligned} \delta &= \frac{1}{64}, & T &= \frac{80}{\rho}, & r &= \frac{\rho\delta^2}{48}, \\ B_0 &= \frac{T}{\delta} + 2, & J &= \lceil \log_2(T/\delta) \rceil + 1, & \varepsilon &= \frac{r}{2B_0^J}, \\ L_{\mathrm{jit}}(\rho) &= \min\left\{\frac{1}{8}, \frac{\rho}{48000}, \frac{\rho\varepsilon^2}{8\log(T/\delta)}\right\}. \end{aligned} \tag{3.2}$$

If $\ell \leq L_{\mathrm{jit}}(\rho)$, *there exists* $0 \leq h \leq 128\pi$ *such that, for an independent* $U_h \sim \mathrm{Unif}[-h/2, h/2]$ *(with* $U_0 = 0$*),*

$$\left\| F_{S+\ell U_h}(x) - \Phi(x) - \frac{\kappa}{6}(1 - x^2)\phi(x) \right\|_\infty \leq \rho\ell, \qquad \kappa = \sum_{i=1}^n \mathbb{E}X_i^3, \tag{3.3}$$

and when $h > 0$, *the symmetrized difference measure* (3.4) *satisfies* $g(2\pi/h) \leq r^2/4$.

Proof. Set $Y_i = X_i/\ell$, let Y_i' be an independent copy of Y_i, and define the probability measure on $[-16, 16]$ by

$$\mu(A) = \frac{\ell^2}{2} \sum_{i=1}^n \mathbb{E}\big[(Y_i - Y_i')^2 \mathbf{1}_{\{Y_i - Y_i' \in A\}}\big]. \tag{3.4}$$

Then $g(u) = \ell^2 \sum_{i=1}^n (1 - |\mathbb{E}e^{\mathrm{i}uY_i}|^2)$ and $|\mathbb{E}e^{\mathrm{i}uS/\ell}| \leq e^{-g(u)/(2\ell^2)}$. Applying Lemma 3.1 together with the signed smoothing inequality and low-frequency Edgeworth estimates of Appendix A bounds the left-hand side of (3.3) by

$$12000\ell^2 + \frac{12r\ell}{\delta^2} + \log(T/\delta) e^{-\varepsilon^2/(2\ell^2)} + \frac{20\ell}{T}, \tag{3.5}$$

each of whose four terms is at most $\rho\ell/4$ under (3.2). □

4 From moment stability to two-cluster support confinement

Combining the uniform jitter approximation of Proposition 3.2 with the variational contact equations of Lemma 2.2 and a quantitative stability form of Esseen's moment inequality (1.6) confines the support of every summand of the selected maximizer.

Proposition 4.1 (Two-cluster support reduction)**.** *Let $0 < \lambda \le 1$ and $N \ge 1$, and set*

$$\rho = 10^{-26}\lambda^2, \qquad \eta = 10^{-10}\lambda, \qquad L_* = \min\left\{L_{\mathrm{jit}}(\rho),\, \frac{1}{2\sqrt{N}}\right\}. \tag{4.1}$$

Every attained maximizer from Lemma 2.1 with penalty parameter η and $\ell \le L_*$ can be written, after dividing each summand by ℓ and relabeling, as independent variables

$$Y_i = h_i(B_i - p_i) + U_i \quad (1 \le i \le m), \qquad Z_1, \dots, Z_s, \tag{4.2}$$

where $m \ge N$, $h > 0$, $B_i \sim \mathrm{Bern}(p_i)$, and

$$p_i \in [1/4, 3/4], \qquad |h_i/h - 1| \le \lambda, \qquad \mathbb{E}(U_i \mid B_i) = 0, \qquad |U_i| \le \lambda h, \tag{4.3}$$

$$\mathbb{E}Z_j = 0, \qquad |Z_j| \le \lambda h, \qquad \sum_{j=1}^{s} \operatorname{Var} Z_j \le \lambda m h^2, \tag{4.4}$$

with the pairs (B_i, U_i) $(1 \le i \le m)$ and variables Z_j $(1 \le j \le s)$ mutually independent.

The complete proof of Proposition 4.1 is given in Appendix B. In brief: rounding $Y_i = X_i/\ell$ to its nearest lattice coset of span $h = 2\pi/p$ incurs a total L^2 error bounded by $h^2 g(p)/8 \le (hr/4)^2$. Quantitative stability in Esseen's lattice moment inequality forces h close to $h_{\mathrm{E}} = (p_{\mathrm{E}}^2 + q_{\mathrm{E}}^2)^{-1}$ and the threshold z close to 0. Retaining the same jitter scale h for each one-summand deletion shows that the one-sided smoothing loss $J_i(z - y + h\ell/2) - G_i(z - y)$ is small at every $y \in \operatorname{supp} X_i$, which forces all pairwise distances between support points of Y_i to lie near $h\mathbb{Z}$. The cubic contact polynomial (B.22) bounds $\operatorname{diam}(\operatorname{supp} Y_i) < 7/2 < 2h$, so each summand either has diameter $< h/2$ (and thus lies in a narrow neighborhood of zero, forming the Z_j's) or has diameter $\approx h$ with all support points concentrated near its two endpoints (forming the two-cluster variables $Y_i = h_i(B_i - p_i) + U_i$ with $p_i \in [1/4, 3/4]$).

5 Local comparison near a common-diameter Bernoulli array

Theorem 5.1 (Explicit local comparison for two-cluster arrays)**.** *Set*

$$\mathcal{A} = 38e^{43}, \qquad \lambda = e^{-\mathcal{A}}, \qquad N = \lceil e^{\mathcal{A}} \rceil. \tag{5.1}$$

Every independent array of the form (4.2)–(4.4) with $m \ge N$ satisfies $\mathcal{R} \le C_{\mathrm{E}}$, where $\mathcal{R} = \Delta/\ell$ is computed from the total variance and the sum of third absolute moments of all $m + s$ summands.

Following the two-regime strategy of He and Cheng [5, Section 3], we prove Theorem 5.1 by comparing the array against the common-diameter Bernoulli reference array with the same label probabilities $p_1, \dots, p_m$, handling small and large total residual variance Λ by complementary arguments that are uniform in unequal probabilities p_i, unequal cluster diameters h_i, and the additional small-diameter summands Z_j.

5.1 Normalization and conditional moment estimates

Set $v_i = p_i(1 - p_i)$ and define the variance-weighted mean diameter

$$\bar{h} = \frac{\sum_{i=1}^m v_i h_i}{\sum_{i=1}^m v_i} \in [(1 - \lambda)h, (1 + \lambda)h].$$

Dividing every summand by $\bar{h}$ (and retaining the notation h_i, U_i, Z_j for the rescaled quantities), we write

$$V = \sum_{i=1}^{m} v_i, \qquad \mu = \sum_{i=1}^{m} p_i, \qquad h_i = 1 + a_i, \qquad \sum_{i=1}^{m} v_i a_i = 0,$$

$$K_0 = \sum_{i=1}^{m} B_i, \qquad A = \sum_{i=1}^{m} a_i(B_i - p_i), \qquad U = \sum_{i=1}^{m} U_i + \sum_{j=1}^{s} Z_j, \qquad W = A + U, \qquad S = K_0 + W,$$

$$\Lambda_A = \sum_{i=1}^{m} v_i a_i^2, \qquad \Lambda_U = \sum_{i=1}^{m} \mathbb{E}U_i^2 + \sum_{j=1}^{s} \mathbb{E}Z_j^2, \qquad \Lambda = \Lambda_A + \Lambda_U, \qquad B^2 = V + \Lambda.$$

Here the uncentered sum S has mean μ, $\mathrm{Cov}(K_0, W) = 0$, and with $\varepsilon_0 = \max(\max_i \|U_i\|_\infty, \max_j \|Z_j\|_\infty)$ and $\xi = \Lambda/V$, the hypotheses (4.3)–(4.4) imply

$$\frac{3m}{16} \le V \le \frac{m}{4}, \qquad \max_{1\le i\le m} |a_i| \le 3\lambda, \qquad \varepsilon_0 \le 2\lambda, \qquad \xi \le 8\lambda, \qquad \Lambda \le 2\lambda m. \tag{5.2}$$

Let T_X and κ denote the sums of the absolute and signed third central moments of all $m + s$ summands, and let

$$T_0 = \sum_{i=1}^{m} v_i(p_i^2 + (1-p_i)^2), \qquad \kappa_0 = \sum_{i=1}^{m} v_i(1 - 2p_i)$$

be the corresponding moments of the unit-diameter reference array $(B_1 - p_1, \dots, B_m - p_m)$. Because $\mathbb{E}(U_i \mid B_i) = 0$ and each cluster preserves its sign ($p_i, 1 - p_i \ge 1/4 > 7/32 > \varepsilon_0$), expanding $|(1 + a_i)(B_i - p_i) + U_i|^3$ gives

$$|T_X - T_0|,\ |\kappa - \kappa_0| \le 2\sqrt{m\Lambda_A} + 4\Lambda, \tag{5.3}$$

$$T_X \ge \frac{V}{3}, \qquad \frac{1}{4} \le \frac{T_X}{B^2} \le 2, \qquad B^2 \le m, \qquad \frac{B}{2} \le P := \frac{B^3}{T_X} \le 4B. \tag{5.4}$$

For $m \ge 10^{12}$ and $|k - \mu| \le 6\sqrt{m}$, Appendix C establishes the count-conditioned gap-perturbation bounds

$$|m_k| := |\mathbb{E}(A \mid K_0 = k)| \le 10^4 \frac{\sqrt{\Lambda_A}}{\sqrt{m}}, \tag{5.5}$$

$$|\operatorname{Var}(A \mid K_0 = k) - \Lambda_A| \le 40000 \frac{\Lambda_A}{\sqrt{m}},$$

$$\mathbb{E}[(A - m_k)^4 \mid K_0 = k] \le 10^7 \Lambda_A^2, \tag{5.6}$$

together with the relative conditional label-probability bound

$$\left| \frac{\mathbb{P}(B_i = b \mid K_0 = k)}{\mathbb{P}(B_i = b)} - 1 \right| \le \frac{500}{\sqrt{m}} \qquad (b \in \{0, 1\}). \tag{5.7}$$

For $\Lambda > 0$, define

$$D = \Lambda^2 + \varepsilon_0^2 \Lambda_U, \qquad H = \max\left\{ \sqrt{\Lambda_A},\ \frac{\Lambda}{\sqrt{\Lambda + \varepsilon_0^2}} \right\}. \tag{5.8}$$

Conditioning first on the full vector of labels $(B_1, \dots, B_m)$ and combining (5.5)–(5.7) with the moment interpolation $\mathbb{E}|Y| \geq (\mathbb{E}Y^2)^{3/2}/(\mathbb{E}Y^4)^{1/2}$ yields, for $|k-\mu| \leq 6\sqrt{m}$,

$$\begin{gathered}\mathbb{E}(W \mid K_0 = k) = m_k, \qquad \frac{\Lambda}{2} \leq \operatorname{Var}(W \mid K_0 = k) \leq 2\Lambda, \\ \mathbb{E}[(W-m_k)^4 \mid K_0 = k] \leq 10^8 D, \qquad \mathbb{E}(|W-m_k| \mid K_0 = k) \geq \frac{H}{30000}, \\ \mathbb{E}(W^4 \mid K_0 = k) \leq 10^9 D, \qquad \mathbb{E}W^4 \leq 400D.\end{gathered} \tag{5.9}$$

5.2 Jittered Edgeworth approximation and central threshold reduction

Let J denote the distribution function of S plus an independent $\mathrm{Unif}[-1/2, 1/2]$ variable. For $T \geq 10$ with $\xi(T+1)^2 \leq 1/56$, set

$$\begin{aligned}E_m(T) = \frac{401(1+\log T)}{\sqrt{m}} &+ (1+\log T)\xi(T+1)^2 \\ &+ 60000(1+\log T)\sqrt{m}\, e^{-m/262144} + \frac{56}{T}.\end{aligned} \tag{5.10}$$

In Appendix A we prove that for $z = (x-\mu)/B$,

$$\sqrt{m}\left\| J(x) - \Phi(z) - \frac{\kappa}{6B^3}(1-z^2)\phi(z) \right\|_\infty \leq E_m(T). \tag{5.11}$$

Choosing the explicit parameters

$$a_0 = e^{43}, \qquad \mathcal{C} = \frac{49}{4}a_0, \qquad \mathcal{B} = \frac{25}{2}a_0, \qquad q = e^{-\mathcal{C}}, \qquad T = e^{\mathcal{B}},$$

direct exponent comparison shows that for every $m \geq N = \lceil e^{\mathcal{A}} \rceil$,

$$\xi(T+1)^2 < \frac{1}{56}, \qquad E_m(T) < \frac{q}{1000}, \qquad m^{-1/2} < \frac{q}{1000}, \qquad 200\lambda < \frac{q}{1000}. \tag{5.12}$$

Shifting the uniform jitter by $\pm 1/2$ bounds either normalized one-sided discrepancy at $z = (x-\mu)/B$ by

$$\phi(z)\left\{ \frac{B^2}{2T_X} + \frac{|\kappa|}{6T_X}|1-z^2| \right\} + 4E_m(T) + m^{-1/2}. \tag{5.13}$$

For $|z| \geq 4$, the main term in (5.13) is at most $(3+z^2)\phi(z) < 1/100$, so the normalized discrepancy for $|z| \geq 4$ is less than $1/24 < C_{\mathrm{E}}$. It therefore suffices to consider central thresholds $|z| \leq 4$.

5.3 Regime I: small residual variance ($0 < \Lambda \leq e^{-43}$)

When $\Lambda = 0$, all two-cluster summands have diameter 1 and $U_i = 0, Z_j = 0$, so (1.2) gives $\mathcal{R} \leq C_{\mathrm{E}}$. Suppose now that $0 < \Lambda \leq e^{-43}$. In Appendix C we prove the following one-sided clipping inequality (compare He and Cheng [5, Lemma 3.2]): if $\mathbb{E}Y = 0$, $\mathbb{E}Y^4 \leq 1/160000$, $f(0) = 0$, and $3/4 \leq f' \leq 5/4$ on $[-7/10, 7/10]$, then for $|u| \leq 3/5$,

$$\begin{aligned}\mathbb{P}(Y > u) + f(u) &\geq \frac{3}{8}\mathbb{E}|Y| - 176000\mathbb{E}Y^4, \\ \mathbb{P}(Y < u) - f(u) &\geq \frac{3}{8}\mathbb{E}|Y| - 176000\mathbb{E}Y^4.\end{aligned} \tag{5.14}$$

By Lemma A.1, the count probabilities satisfy

$$b_k := \mathbb{P}(K_0 = k) \geq \frac{e^{-100}}{\sqrt{m}} \qquad (|k-\mu| \leq 6\sqrt{m}). \tag{5.15}$$

Write any threshold with $|z| \leq 4$ as $x = k + u$, where k is a nearest integer to x and $|u| \leq 1/2$, so that $|k-\mu| \leq 5\sqrt{m}$. Setting $f_k(u) = [\Phi((k-\mu+u)/B) - \Phi((k-\mu)/B)]/b_k$, we have $3/4 \leq f_k' \leq 5/4$ on $[-1,1]$. Decomposing the event $\{S \leq k+u\}$ by the value of K_0 gives

$$\begin{aligned} F_S(k+u) &= F_{K_0}(k) - b_k \mathbb{P}(W > u \mid K_0 = k) + e^+_{k,u}, \\ F_S^-(k+u) &= F_{K_0}(k-1) + b_k \mathbb{P}(W < u \mid K_0 = k) + e^-_{k,u}, \end{aligned} \tag{5.16}$$

where Markov's inequality and (5.9) bound the off-layer tail sums by

$$|e^{\pm}_{k,u}| \leq \sum_{j \neq k} b_j \mathbb{P}\{|W| \geq |j-k| - 1/2 \mid K_0 = j\} \leq 2 \cdot 10^{11} \frac{D}{\sqrt{m}}.$$

Applying (5.14) conditionally on $K_0 = k$ to $Y = W - m_k$ (with $10^8 D \leq 2 \cdot 10^8 e^{-86} < 1/160000$) and using $Pb_k \geq \frac{2}{5}\phi(4.01) > e^{-11}$ and $(3/8)(H/30000)e^{-11} > e^{-23}H$, while comparing $P = B^3/T_X$ to $V^{3/2}/T_0$ via (5.3), bounds either normalized discrepancy branch at $x = k+u$ by

$$R_0 - e^{-23}H + e^{40}\left\{D + \frac{\sqrt{\Lambda_A} + \Lambda}{\sqrt{m}}\right\} < R_0 \leq C_{\mathrm{E}}, \tag{5.17}$$

where $R_0 = \mathcal{R}(B_1 - p_1, \ldots, B_m - p_m) \leq C_{\mathrm{E}}$ by (1.2) and the error bracket in (5.17) is strictly smaller than $e^{-23}H$ since $D/H \leq (\Lambda + \varepsilon_0^2)^{3/2} \leq 3e^{-129/2}$ and $m \geq e^{\mathcal{A}}$.

5.4 Regime II: larger residual variance ($\Lambda \geq e^{-43}$)

Suppose now that $\Lambda \geq e^{-43}$ and $|z| \leq 4$. For every integer k with $|k - x| \leq \max(1, \sqrt{\Lambda})$ (at least $\sqrt{\Lambda}$ integers, all satisfying $|k-\mu| \leq 6\sqrt{m}$), tilt the Bernoulli probabilities exponentially as in Appendix C so that the tilted count mean equals k, and project A orthogonally to $K_0 - k$. Conditional on $K_0 = k$, we have $W = m_k^0 + R_k$, where under the tilted measure R_k is a sum of independent centered pair residuals and Z_j's satisfying

$$L := \mathrm{Var}_\theta\, R_k \in [\Lambda/2, 2\Lambda], \qquad \mathrm{Var}_\theta\, K_0 \geq \frac{m}{10}, \qquad |m_k^0| \leq \sqrt{\Lambda}, \qquad \mathrm{Cov}_\theta(K_0, R_k) = 0. \tag{5.18}$$

Applying Lemmas A.2 and A.3 to both quarter-intervals $(1/4, 1/2)$ and $(-1/2, -1/4)$ with $L_0 = \frac{1}{2}e^{-43}$, $C_0 = 4 + 2e^{43/2}$, $A_g = 4e^{43/2}$, $\omega = 16e^{A_g^2/6}$, and $T_g = 4\omega$ shows that conditional on $K_0 = k$, $R_k - (x - k - m_k^0)$ falls in each quarter-interval with probability at least $\phi(A_g)/(8\omega\sqrt{L})$. Multiplying by $b_k \geq e^{-100}/\sqrt{m}$ and summing over the chosen integers k yields

$$\mathbb{P}\{S - x \in (1/4, 1/2)\} \geq \frac{q}{\sqrt{m}}, \qquad \mathbb{P}\{S - x \in (-1/2, -1/4)\} \geq \frac{q}{\sqrt{m}}. \tag{5.19}$$

Consequently, the one-sided jitter smoothing losses satisfy $J(x + 1/2) - F_S(x) \geq q/(2\sqrt{m})$ and $F_S^-(x) - J(x - 1/2) \geq q/(2\sqrt{m})$. Multiplying by $P \geq B/2 \geq \sqrt{3m}/8$ subtracts at least $q/10$ from the main term of (5.13) (which is at most $C_{\mathrm{E}} + 200\lambda$ by Esseen's Bernoulli inequality $3V + |\kappa_0| \leq c_{\mathrm{E}} T_0$ and (5.3)), bounding either branch below $C_{\mathrm{E}} - q/10 + 200\lambda + 4E_m(T) + m^{-1/2} < C_{\mathrm{E}} - q/20$. This completes the proof of Theorem 5.1.

6 Composition of constants and proof of Theorem 1.1

Take $\mathcal{A} = 38e^{43}$, $\lambda = e^{-\mathcal{A}}$, and $N = \lceil e^{\mathcal{A}} \rceil$ from (5.1), and define ρ, η, L_* by (4.1). If an independent array violated $\Delta \le C_{\mathrm{E}}\ell$ at some Lyapunov ratio $\ell \le t_*$, where

$$t_* := \min\left\{\frac{L_*}{2}, \frac{\eta^3}{108}, \frac{1}{2000}\right\}, \tag{6.1}$$

then Lemma 2.1 would produce an attained maximizer with $\ell < 2t_* \le L_*$ and ratio $R > C_{\mathrm{E}}$. Proposition 4.1 would then place that maximizer in the class (4.2)–(4.4) with $m \ge N$, contradicting Theorem 5.1.

It remains only to verify that $\ell_0 = \exp(-\exp(97)) < t_*$. Put $A = \mathcal{A} = 38e^{43} > 1000$, so $\lambda = e^{-A}$, $N = \lceil e^A \rceil$, $\rho = 10^{-26}e^{-2A}$, and $\eta = 10^{-10}e^{-A}$. In (3.2), with $Q = T/\delta = 5120/\rho$, $B_0 = Q + 2$, $J = \lceil \log_2 Q \rceil + 1$, $r = \rho/(48 \cdot 4096)$, and $\varepsilon = r/(2B_0^J)$, the bounds $\log 2 > 2/3$ and $\log 10 < 3$ give

$$\log(1/\rho),\ \log Q,\ \log B_0,\ \log(1/r) < 3A, \qquad J < \frac{\log Q}{\log 2} + 2 < 5A,$$

and hence $\log(1/\varepsilon) < 3A + 1 + 15A^2 < 16A^2$. Consequently,

$$-\log L_{\mathrm{jit}}(\rho) \le \log(1/\rho) + 2\log(1/\varepsilon) + \log(8\log Q) < 32A^2 + 4A < 40A^2.$$

Since $\log(2\sqrt{N}) < A$ and $3A + 30\log 10 + \log 108 < 4A$, we obtain

$$-\log t_* < 41A^2 = 41 \cdot 38^2 e^{86} = 59204e^{86}.$$

Using $e > 163/60$ and $(163/60)^{11} > 59204$ gives

$$-\log \ell_0 = e^{97} > 59204e^{86} > -\log t_*,$$

so $\ell_0 < t_*$. This completes the proof of Theorem 1.1.

A Quantitative Fourier and smoothing estimates

Write $\widehat{\nu}(t) = \int_{\mathbb{R}} e^{itx}\,\nu(dx)$ and $\operatorname{sinc} u = (\sin u)/u$ (with $\operatorname{sinc} 0 = 1$). This appendix establishes the explicit signed-smoothing, low-frequency Edgeworth, Poisson-binomial count, and conditional Fourier bounds used in Sections 3 and 5.

A.1 Signed smoothing and low-frequency Edgeworth expansion

Let F be a probability distribution function with finite first absolute moment, and let G be the distribution primitive of a finite signed measure of total mass one, finite first absolute moment, and Lipschitz constant M_G. By the signed smoothing inequality of He and Cheng [5, Lemma 2.1 and equation (71)], for every $A > 0$,

$$\|F - G\|_\infty \le \frac{1}{4}\int_{-A}^{A} \frac{|\widehat{F}(t) - \widehat{G}(t)|}{|t|}\,dt + \frac{24M_G}{A}. \tag{A.1}$$

(Indeed, convolving with the shifted Fejér-type kernel $k(x) = 3\operatorname{sinc}(x/4)^4/(8\pi)$, whose Fourier transform is supported on $[-1,1]$ and which satisfies $\varepsilon_k = \int_{16}^{\infty} k(x)\,dx < 1/8$ and $\int_{-\infty}^{16}(16-x)k(x)\,dx < 18$,

gives coefficients $1/[2\pi(1-2\varepsilon_k)] \le 1/4$ and $18/(1-2\varepsilon_k) \le 24$; in particular, the weaker coefficients $1/\pi$ and 40 also hold.)

In the setting of Proposition 3.2, with $v_i = \mathbb{E}X_i^2$, $\kappa = \sum_{i=1}^n \mathbb{E}X_i^3$, and $f_i(t) = \mathbb{E}e^{\mathrm{i}tX_i}$, we have on $|t| \le 1/(64\ell)$ that $|f_i(t) - 1| \le v_i t^2/2 \le 1/128$, $\sum_i \mathbb{E}X_i^4 \le 8\ell^2$, and $\sum_i v_i^2 \le 64\ell^2$. Expanding $\sum_i \log f_i(t) = -t^2/2 + \kappa(\mathrm{i}t)^3/6 + w(t)$ with $|w(t)| \le (49/3)\ell^2 t^4 \le t^2/6$ bounds the difference against $\widehat{\nu}(t) = e^{-t^2/2}(1 + \kappa(\mathrm{i}t)^3/6)$ by

$$|\widehat{F}(t) - \widehat{\nu}(t)| \le \ell^2 e^{-t^2/3} \left\{ \frac{49}{3} t^4 + \frac{t^6}{72} \right\}. \tag{A.2}$$

Including the jitter factor $\mathrm{sinc}(h\ell t/2)$, integrating over $|t| \le 1/(64\ell)$ for $h \le 128\pi$, adding the Gaussian tail beyond $a = 1/(64\ell)$, and applying Lemma 3.1 with the smoothing constants $(1/\pi, 40)$ and $M_G \le 1/2$ yields (3.5).

For the local comparison in Section 5, telescoping the product directly against e^{A_i} with $A_i = -v_i t^2/2 + \mathbb{E}X_i^3(\mathrm{i}t)^3/6$ gives the tighter low-frequency bound

$$\int_{|t| \le 1/(64\ell)} \frac{|\,\mathrm{sinc}(h\ell t/2)\widehat{F}(t) - \widehat{\nu}(t)|}{|t|}\, dt \le \left(73 + \frac{h^2}{8}\right) \ell^2. \tag{A.3}$$

A.2 Local Gaussian approximation for Poisson-binomial counts

Lemma A.1. *Let $K = \sum_{i=1}^m B_i$ be a sum of independent Bernoulli variables with mean $\mu = \mathbb{E}K$ and variance $V = \mathrm{Var}\, K \ge 1/2$. Then for every $k \in \mathbb{Z}$,*

$$\left| \mathbb{P}(K = k) - \frac{\phi((k-\mu)/\sqrt{V})}{\sqrt{V}} \right| \le \frac{6}{V}, \qquad \sup_{k \in \mathbb{Z}} \mathbb{P}(K = k) < \frac{1}{\sqrt{V}}. \tag{A.4}$$

In particular, if $p_i \in [\alpha, 1-\alpha]$ for all i, $\beta = \alpha(1-\alpha)$, $\gamma = \phi(M/\sqrt{\beta})$, and $m \ge 144/(\beta\gamma^2)$, then $\mathbb{P}(K = k) \ge \gamma/\sqrt{m}$ whenever $|k - \mu| \le M\sqrt{m}$.

Proof. For $Z_i = B_i - p_i$ and $v_i = p_i(1-p_i)$, on $t \in [-\pi, \pi]$ we have $|\mathbb{E}e^{\mathrm{i}tZ_i}| \le e^{-2v_i \sin^2(t/2)} \le e^{-2v_i t^2/\pi^2}$ and $|\mathbb{E}e^{\mathrm{i}tZ_i} - e^{-v_i t^2/2}| \le v_i(|t|^3/6 + t^4/32)$. Telescoping the product with deleted decay $e^{-(V-v_i)t^2/\pi^2} \le e^{-Vt^2/\pi^2}$ and integrating over $[-\pi, \pi]$ gives an error at most $(\pi^3/12 + \pi^4/64 + 1/\pi^2)/V < 6/V$. □

A.3 Conditioning an independent product on a Bernoulli count

Lemma A.2. *Let (B_i, R_i) $(1 \le i \le m)$ be independent pairs with $B_i \sim \mathrm{Bern}(r_i)$, $r_i \in [a, 1-a]$, $\mathbb{E}R_i = 0$, and $|R_i| \le d_0$, and let Z_j $(1 \le j \le s)$ be additional independent centered variables with $|Z_j| \le d_0$. Put*

$$K = \sum_{i=1}^m B_i, \qquad R = \sum_{i=1}^m R_i + \sum_{j=1}^s Z_j, \qquad \mathbb{E}K = k \in \mathbb{Z}, \qquad V = \mathrm{Var}\, K \ge 1000, \qquad L = \mathrm{Var}\, R > 0,$$

and assume $\mathrm{Cov}(K, R) = 0$. Set $\beta = a(1-a)$. If $g : \mathbb{R} \to \mathbb{R}$ is continuous and integrable with integrable Fourier transform supported on $[-T, T]$, and $d_0 T \le \beta/162$, then for $Z \sim N(0,1)$,

$$\begin{aligned} &\sup_{y \in \mathbb{R}} |\mathbb{E}[g(R - y) \mid K = k] - \mathbb{E}g(\sqrt{L}Z - y)| \\ &\qquad \le \frac{\|g\|_1}{\sqrt{L}} \left\{ \frac{32}{\sqrt{V}} + \frac{8d_0}{\sqrt{L}} + 2T\sqrt{VL}\, e^{-\beta m/162} \right\}. \end{aligned} \tag{A.5}$$

Proof. Write $\chi(s,t) = \mathbb{E}e^{\mathrm{i}s(K-k)+\mathrm{i}tR}$. On $|s| \le 1/4$ and $|t| \le T$, each centered phase $X = s(B_i - r_i) + tR_i$ or tZ_j satisfies $|X| \le 1/2$, $|\mathbb{E}e^{\mathrm{i}X}| \le e^{-\mathbb{E}X^2/3}$, and $|\mathbb{E}e^{\mathrm{i}X} - e^{-\mathbb{E}X^2/2}| < \mathbb{E}|X|^3/4$. Since $\mathrm{Cov}(K,R) = 0$, the total phase variance is $Vs^2 + Lt^2$ and the sum of third absolute moments is at most $4(V|s|^3 + d_0L|t|^3)$, so telescoping gives

$$|\chi(s,t) - e^{-Vs^2/2 - Lt^2/2}| \le 2(V|s|^3 + d_0L|t|^3)e^{-(Vs^2+Lt^2)/3},$$

whose double integral over $|s| \le 1/4$, $|t| \le T$ is at most $56(VL)^{-1/2}(V^{-1/2} + d_0L^{-1/2})$. On $1/4 \le |s| \le \pi$, each Bernoulli factor has modulus at most $1 - \beta/81$ and the residual perturbation changes it by at most $Td_0 \le \beta/162$, giving $|\chi(s,t)| \le e^{-\beta m/162}$. Dividing the joint Fourier inversion formula by $\mathbb{P}(K = k) \ge \frac{1}{2}(2\pi V)^{-1/2}$ (from Lemma A.1 since $V \ge 1000$) yields (A.5). □

Lemma A.3 (Bandlimited interval minorant)**.** *For an interval (a,b) with midpoint $w_0 = (a+b)/2$ and half-width $d = (b-a)/2$, fix $L_0 > 0$, $C_0 \ge 0$, and choose*

$$A_g \ge C_0 + \frac{|w_0| + d/2}{\sqrt{L_0}}, \qquad \omega = \frac{2}{d}e^{A_g^2/6}, \qquad T = 4\omega.$$

Then the function

$$g(w) = \left(1 - \frac{(w-w_0)^2}{d^2}\right)\mathrm{sinc}\big(\omega(w-w_0)\big)^4$$

satisfies $g \le \mathbf{1}_{(a,b)}$, has integrable Fourier transform supported on $[-T,T]$, and obeys

$$\|g\|_1 \le \frac{4}{\omega}, \qquad \mathbb{E}g(\sqrt{L}Z - y) \ge \frac{\phi(A_g)}{4\omega\sqrt{L}} \quad (L \ge L_0,\ |y| \le C_0\sqrt{L}).$$

Consequently, if the bracket in (A.5) is at most $\phi(A_g)/32$, then $\mathbb{P}(R - y \in (a,b) \mid K = k) \ge \phi(A_g)/(8\omega\sqrt{L})$.

Proof. On $|w - w_0| \le 1/\omega \le d/2$, we have $g(w) \ge (3/4)(5/6)^4 > 1/4$ and Gaussian density at least $\phi(A_g)/\sqrt{L}$, while the negative part of g outside (a,b) has integral at most $2/(d^3\omega^4) = \phi(A_g)/(4\omega\phi(0))$ and Gaussian density at most $\phi(0)/\sqrt{L}$; subtracting gives the lower bound on $\mathbb{E}g(\sqrt{L}Z - y)$. □

A.4 Jitter on the rescaled count lattice

To establish (5.11) in Section 5 (compare He and Cheng [5, Lemma A.4]), set $g(t) = \sum_i(1 - |f_i(t)|^2)$ (over all $m+s$ summands) and $g_0(t) = 2V(1 - \cos t)$. Because $\sum_{i=1}^m v_i a_i = 0$ and $\mathbb{E}(U_i \mid B_i) = 0$, the first-order terms in the Taylor expansion of $1 - \cos(td)$ around the unperturbed differences $D_0 = B_i - B_i'$ vanish, giving for $\max_i |a_i|, \varepsilon_0 \le 1/4$:

$$\begin{aligned} |g(t) - g_0(t)| &\le \Lambda t^2, \\ |g'(t) - g_0'(t)| &\le \Lambda(2|t| + 2t^2), \\ |g''(t) - g_0''(t)| &\le \Lambda(2 + 8|t| + 4t^2). \end{aligned} \tag{A.6}$$

When $T \ge 10$ and $\xi(T+1)^2 \le 1/56$, each resonance interval $I_j = [2\pi j - 1/4, 2\pi j + 1/4]$ intersecting $[-T,T]$ contains a unique minimum t_j of g with $|t_j - 2\pi j| \le 4\Lambda(T+1)^2/V$ and $\prod_i |f_i(t)| \le e^{-V(t-t_j)^2/4}$ on I_j, while outside $\bigcup_j I_j$ we have $\prod_i |f_i(t)| \le e^{-V/50}$. Since $\mathrm{sinc}(t/2)$ vanishes at $2\pi j$ and is $(1/4)$-Lipschitz, summing over $j \ne 0$ gives

$$\sum_{j \ne 0} \int_{I_j \cap [-T,T]} \frac{|\mathrm{sinc}(t/2)\prod_i f_i(t)|}{|t|}\,dt \le (1 + \log T)\left\{\frac{2}{5V} + \frac{4\sqrt{\pi}}{5}\frac{\Lambda(T+1)^2}{V\sqrt{V}}\right\}. \tag{A.7}$$

Combining (A.3) on $|u| \le 1/128$, (A.7) on the nonzero resonance intervals, the exponential decay $e^{-V/50}$ outside $\bigcup_j I_j$, and the smoothing bound (A.1) at cutoff BT yields (5.11) with error (5.10).

B Quantitative confinement of the summand supports

In this appendix we prove Proposition 4.1. We first establish a quantitative stability refinement of Esseen's lattice moment inequality and a two-shift smoothing lemma, and then apply them to the variational contact equations of Lemma 2.2.

B.1 Quantitative stability in Esseen's moment inequality

Throughout this subsection we write $c = c_{\mathrm{E}}$, $p = p_{\mathrm{E}}$, and $\tau = \tau_{\mathrm{E}}$ (compare He and Cheng [5, Lemma A.9]). Let Y be a centered, nondegenerate random variable supported on a coset of $h\mathbb{Z}$ ($h > 0$). Its oriented Esseen moment deficit is

$$\delta_Y = \frac{c\mathbb{E}|Y|^3 - \mathbb{E}Y^3 - 3h\mathbb{E}Y^2}{h\mathbb{E}Y^2}.$$

Decomposing $\mathrm{Law}(Y)$ into a mixture of δ_0 and centered two-point laws $P_{u,v}$ on $\{-u, v\}$ ($u, v > 0$) via the measure $\lambda(du, dv) = (u+v)P_-(du)P_+(dv)/\mathbb{E}Y_+$ and passing to the variance-weighted probability measure $\omega(du, dv) = uv\lambda(du, dv)/\mathbb{E}Y^2$, let $n = (u+v)/h \in \{1, 2, \ldots\}$ and $a = u/(u+v) \in (0, 1)$. The algebraic identity $c\{a^2 + (1-a)^2\} - (1-2a) = 3 + 2c(a-p)^2$ yields the exact representation

$$\delta_Y = \int \{3(n-1) + 2cn(a-p)^2\}\, d\omega. \tag{B.1}$$

In particular $\delta_Y \ge 0$, which recovers (1.6).

When $\delta_Y < 3$, some pair of support points has $n = 1$; these must be the unique two lattice points $-ah, (1-a)h$ of the coset straddling 0, and they carry ω-weight at least $1 - \delta_Y/3$. Since every other two-point component has variance at least $h^2 a(1-a)$, $\mathrm{Law}(Y)$ decomposes as

$$\begin{aligned} \mathrm{Law}(Y) &= (1 - r_Y)\,\mathrm{Law}\big(h(B_a - a)\big) + r_Y Q, \qquad r_Y \le \frac{\delta_Y}{3}, \\ |a - p| &\le \sqrt{\frac{\delta_Y}{2c(1 - \delta_Y/3)}}, \qquad h^2 a(1-a) \le \mathbb{E}Y^2 \le \frac{h^2 a(1-a)}{1 - \delta_Y/3}, \end{aligned} \tag{B.2}$$

and if $|Y| \le M$, then $W_1(\mathrm{Law}(Y), \mathrm{Law}(h(B_p - p))) \le \frac{2M}{3}\delta_Y + 2h\sqrt{\delta_Y/[2c(1-\delta_Y/3)]}$. Summing over a family of centered h-lattice variables Y_i with $V = \sum_i \mathbb{E}Y_i^2$, $B = \sum_i \mathbb{E}|Y_i|^3$, $K = \sum_i \mathbb{E}Y_i^3$, and aggregate deficit $\Delta_L = (cB - K - 3hV)/(hV)$ gives, by Cauchy–Schwarz,

$$\frac{1}{V}\sum_i \mathbb{E}Y_i^2 W_1(\mathrm{Law}(Y_i), \mathrm{Law}(h(B_p - p))) \le \left(\frac{5M}{3} + h\right)\Delta_L + h\sqrt{\frac{3}{c}}\sqrt{\Delta_L}, \tag{B.3}$$

and if $\Delta_L \le D$, then with $a_D = \sqrt{(1 + D/3)D/(2c)}$, $b_D = \tau D/3 + D/c + (2/c)a_D$, and $k_D = D/(3c) + 2a_D$,

$$\left|\frac{B}{hV} - \tau\right| \le b_D, \qquad \left|\frac{K}{hV} - \frac{1}{c}\right| \le k_D, \qquad \frac{B}{hV} \ge \frac{1}{2}. \tag{B.4}$$

B.2 Rounding to a lattice coset at the selected period

For the selected maximizer of Lemma 2.1, set $Y_i = X_i/\ell$, so $|Y_i| \le 8$, $\mathbb{E}Y_i^2 \le 9$, and $\ell^2 \sum_i \mathbb{E}Y_i^2 = \ell^2 \sum_i \mathbb{E}|Y_i|^3 = 1$. At a frequency $u \neq 0$ with $h = 2\pi/|u|$, round each Y_i to the nearest point of the lattice coset aligned with $\arg \mathbb{E}e^{\mathrm{i}uY_i}$ and center the rounded variable to obtain Q_i. Using $|v| \le (\pi/2)|e^{\mathrm{i}v} - 1|$ on $[-\pi, \pi]$, we have

$$e := \ell^2 \sum_{i=1}^{n} \mathbb{E}|Y_i - Q_i|^2 \le \frac{h^2 g(u)}{8}, \qquad |Q_i| \le 8 + h =: M, \tag{B.5}$$

and Cauchy–Schwarz bounds the moment differences by

$$\begin{aligned} \ell^2 \sum_{i=1}^{n} |\mathbb{E}Q_i^2 - \mathbb{E}Y_i^2| &\le 2\sqrt{e} + e, \\ \ell^2 \sum_{i=1}^{n} \big|\mathbb{E}|Q_i|^3 - \mathbb{E}|Y_i|^3\big|,\ \ell^2 \sum_{i=1}^{n} |\mathbb{E}Q_i^3 - \mathbb{E}Y_i^3| &\le \frac{3M}{2}\sqrt{e}(2 + \sqrt{e}). \end{aligned} \tag{B.6}$$

B.3 Support spacing from two small smoothing losses

Lemma B.1. *Let F be the distribution function of a finite positive measure ν, and let $J = F * \mathrm{Unif}[-h/2, h/2]$ $(h > 0)$. Suppose $J(x)$ differs from an affine function of slope $q > 0$ by at most ε on $[-D_0 - 2h, 3h]$. Put $H(a) = J(a + h/2) - F(a) = \int_{(a,a+h)} (1 - (t-a)/h)\, \nu(dt) \ge 0$. If $0 \le d \le D_0$ and $H(0), H(-d) \le s$, then for any $0 < r_0 \le h/4$ and $K_0 = \lceil D_0/h \rceil$,*

$$\mathrm{dist}(d, h\mathbb{Z}) \le 3r_0 + \frac{4s + (5 + 4K_0)\varepsilon + 4K_0 h\varepsilon/r_0}{q}. \tag{B.7}$$

Proof. For $0 < d < h$, the trapezoid kernel $T_d(t) = h^{-1} \int_{h/2-d}^{h/2-d/2} \mathbf{1}_{\{|x-t|<h/2\}}\, dx$ satisfies $\min\{1, 2(h-d)/d\} T_d(t) \le w_0(t) + w_{-d}(t)$ pointwise, where $w_a(t) = (1 - (t-a)/h)\mathbf{1}_{(a,a+h)}(t)$. Integrating against ν and using $J(h/2 - d/2) - J(h/2 - d) \ge qd/2 - 2\varepsilon$ gives

$$q \min(d/2, h - d) \le H(0) + H(-d) + 2\varepsilon. \tag{B.8}$$

For general $d = kh + d_0$ $(0 \le k \le K_0,\ 0 \le d_0 < h)$, averaging H over an interval of length r_0 and using $|\overline{H}_{r_0}(a + h) - \overline{H}_{r_0}(a)| \le 2\varepsilon(1 + h/r_0)$ transfers the bound $H(-d) \le s$ across k periods to $-d_0$, where (B.8) yields (B.7). □

B.4 Pinning the lattice span h and threshold z

For the selected maximizer, put $v_i = \mathbb{E}Y_i^2$, $M_i = \mathbb{E}[Y_i|Y_i|]$, and $\theta = \ell^{-1} \sum_i \mathbb{E}X_i^3 \in [-1, 1]$. Let $\rho, r, T, \varepsilon, h$ be as in Proposition 3.2, and define the explicit error quantities

$$\begin{aligned} u &= \rho + \ell/2, & D &= 16u + 256r, \\ H_h &= 72r + 5b_D, & Z^2 &= 10(u + 18r), \\ S_0 &= 2\rho + 18r + 460\ell^2, & A_0 &= 2\rho + 24\{4\ell^2 + (Z + 32\ell)^2/2 + \ell/5\}, \\ \gamma &= 300\sqrt{A_0} + 11S_0 + 131A_0, & W_0 &= 21D + 3\sqrt{D} + 42r, \\ E_P &= 14000(u + 18r + \eta) + 1100Z^2 + 4000\ell + 4000\ell^2 + 1100\eta, \\ E_{\mathrm{end}} &= E_P + 6100\gamma + 700\gamma^2 + 500H_h, \end{aligned} \tag{B.9}$$

which under (4.1) satisfy

$$r \le 10^{-6}, \qquad D \le 10^{-3}, \qquad A_0 \le \frac{9}{64}, \qquad \gamma \le \frac{1}{100}, \\ E_P \le 10^{-4}, \qquad E_{\mathrm{end}} \le \frac{1}{2000}, \qquad W_0 \le \frac{1}{4}. \tag{B.10}$$

Since $h = 0$ would give $R \le \phi(0)/6 + \rho < C_{\mathrm{E}}$ by (3.3), we have $h > 0$. Applying (B.5)–(B.6) first with $h \le 128\pi$ gives $h < 5/2$, and reapplying them with $h < 5/2$ gives

$$\sqrt{e} \le r, \qquad |V_Q - 1| \le 3r, \qquad |B_Q - 1| \le 32r, \qquad |K_Q - \theta| \le 32r, \tag{B.11}$$

so Esseen's inequality (1.6) for Q_i implies

$$3h + |\theta| \le c + 256r. \tag{B.12}$$

Conversely, monotonicity $F_S(z) \le F_{S+\ell U_h}(z + h\ell/2)$ and (3.3) give

$$R \le \phi(z)\left\{\frac{h}{2} + \frac{\theta}{6}(1 - z^2)\right\} + u, \tag{B.13}$$

which together with $R > C_{\mathrm{E}}$ and (B.12) implies

$$3h + |\theta| > c - 16u, \qquad R \le C_{\mathrm{E}} + u + 18r. \tag{B.14}$$

Thus the aggregate Esseen deficit of (Q_i) is at most D, so (B.4) yields

$$|h - h_{\mathrm{E}}| \le H_h < \frac{1}{40}, \qquad \big||\theta| - 1/\sqrt{10}\big| \le 34r + 3k_D + \frac{H_h}{6} < \frac{1}{10}, \tag{B.15}$$

and hence

$$\frac{19}{10} < h < 2, \qquad \theta > \frac{1}{5}, \tag{B.16}$$

while $C_{\mathrm{E}} < (C_{\mathrm{E}} + 18r)e^{-z^2/2} + u$ gives

$$|z| \le Z, \qquad 0 \le K - C_{\mathrm{E}} \le u + 18r + \eta. \tag{B.17}$$

B.5 Deleted jitter and pairwise support spacing

Deleting summand i reduces $g(u)$ by at most ℓ^2, so the deleted sum $S - X_i$ convolved with the same jitter ℓU_h has distribution function J_i satisfying, with $\sigma_i^2 = 1 - s_i \ge 3/4$, $\kappa_- = \sum_{j \ne i} \mathbb{E}X_j^3$, and $G_i^*(x) = \Phi(x/\sigma_i) + \frac{\kappa_-}{6\sigma_i^3}(1 - (x/\sigma_i)^2)\phi(x/\sigma_i)$,

$$\|J_i - G_i^*\|_\infty \le 6000\ell^2 + \frac{24r\ell}{\delta^2} + 2\log(T/\delta)e^{-\varepsilon^2/(2\ell^2)} + \frac{20\ell}{T} \le 2\rho\ell. \tag{B.18}$$

At any support point $y = \ell t \in \operatorname{supp} X_i$, the contact equality $D_i(y) = 0$ and (2.6) give

$$|G_i(z - y) - \Phi((z - y)/\sigma_i) - R\ell| \le 450\ell^3, \tag{B.19}$$

which combined with (B.12) and (B.18) bounds the one-sided smoothing loss by

$$0 \le \frac{J_i(z - y + h\ell/2) - G_i(z - y)}{\ell} \le S_0. \tag{B.20}$$

Applying Lemma B.1 to $a \mapsto \ell^{-1}J_i(z - x + \ell a)$ around any $x \in \operatorname{supp} X_i$ shows that every pair $x, y \in \operatorname{supp} X_i$ satisfies

$$\operatorname{dist}((y - x)/\ell, h\mathbb{Z}) \le \gamma. \tag{B.21}$$

B.6 The cubic contact polynomial and two-cluster decomposition

Dividing (2.8) by $\phi(0)\ell^3/6$ and using (2.7) and (B.17) shows that every $t \in \operatorname{supp} Y_i$ satisfies

$$P_i(t) := t^3 - c|t|^3 + \frac{3c}{2}t^2 + 3(cM_i - v_i)t \geq -E_P. \tag{B.22}$$

For the extreme points $-a, b$ of $\operatorname{supp} Y_i$, combining $P_i(-a), P_i(b) \geq -E_P$ with $v_i \leq ab$ and $|M_i| \leq v_i$ gives $a + b < 7/2 < 2h$.

Consequently, by (B.21), if $a+b < h/2$, then $a+b \leq \gamma$ and $|Y_i| \leq \gamma$; we relabel these summands as $Z_1, \ldots, Z_s$. If $a+b \geq h/2$, then $|(a+b) - h| \leq \gamma$ and every point of $\operatorname{supp} Y_i$ lies within 2γ of $-a$ or b. Conditioning on the indicator B_i of the upper cluster $[(-a+b)/2, b]$ yields

$$Y_i = h_i(B_i - p_i) + U_i, \qquad \mathbb{E}(U_i \mid B_i) = 0, \qquad |h_i - h| \leq 5\gamma, \qquad |U_i| \leq 2\gamma, \tag{B.23}$$

and substituting the cluster endpoints into (B.22) forces $p_i \in [1/4, 3/4]$. Finally, (B.3) and the rounding bound (B.11) give

$$\sum_{i=1}^{n} \ell^2 v_i W_1(\operatorname{Law}(Y_i), \operatorname{Law}(h(B_{p_{\mathrm{E}}} - p_{\mathrm{E}}))) \leq W_0, \tag{B.24}$$

so the small-diameter summands Z_j carry total original variance at most $W_0/(2hp_{\mathrm{E}}q_{\mathrm{E}} - \gamma) \leq 2W_0$, while the two-cluster summands have count m satisfying

$$\frac{1 - 2W_0}{2\ell^2} \leq m \leq \frac{2}{\ell^2}, \qquad m \geq \frac{1}{4\ell^2} \geq N, \tag{B.25}$$

and $\sum_{j=1}^{s} \operatorname{Var} Z_j/(mh^2) < 3W_0 \leq \lambda$. Checking the numerical gates (B.10) under (4.1) completes the proof of Proposition 4.1.

C Conditional moment bounds and the one-sided clipping inequality

This appendix proves the count-conditioned moment estimates (5.5)–(5.7) and the one-sided clipping inequality (5.14) used in Section 5.

C.1 Exponential tilting and orthogonal projection

Let $p_i \in [1/4, 3/4]$, $v_i = p_i(1 - p_i)$, $\sum_{i=1}^{m} v_i a_i = 0$, $\Lambda_A = \sum_{i=1}^{m} v_i a_i^2 > 0$, $m \geq 10^{12}$, and $|k - \mu| \leq 6\sqrt{m}$. Choose the unique tilt parameter $\theta \in [-\log 2, \log 2]$ so that the tilted Bernoulli variables $B_i \sim \operatorname{Bern}(r_i)$ with

$$r_i = \frac{p_i e^{\theta}}{1 - p_i + p_i e^{\theta}} \in \left[\frac{1}{8}, \frac{7}{8}\right], \qquad w_i = r_i(1 - r_i) \geq \frac{7}{64} > \frac{1}{10}, \qquad V_\theta = \sum_{i=1}^{m} w_i$$

satisfy $\sum_{i=1}^{m} r_i = k$. Then

$$|\theta| \leq \frac{60}{\sqrt{m}}, \qquad \left|\frac{w_i}{v_i} - 1\right| \leq 2|\theta|, \qquad |r_i - p_i - \theta v_i| \leq \theta^2 v_i. \tag{C.1}$$

Because the exponential factor $e^{\theta K_0}$ is constant on the event $\{K_0 = k\}$, the conditional distribution given $K_0 = k$ is identical under the original and tilted measures. Setting

$$c_\theta = \frac{\sum_{i=1}^m w_i a_i}{V_\theta}, \qquad b_i = a_i - c_\theta, \qquad L = \sum_{i=1}^m b_i(B_i - r_i), \qquad Q = \sum_{i=1}^m b_i^2, \qquad Q_\theta = \sum_{i=1}^m w_i b_i^2,$$

we have $A = m_k^0 + L$ on $\{K_0 = k\}$, where $\mathrm{Cov}_\theta(K_0, L) = \sum_{i=1}^m w_i b_i = 0$ and

$$|m_k^0| = \left|\sum_{i=1}^m a_i(r_i - p_i)\right| \le 1800\frac{\sqrt{\Lambda_A}}{\sqrt{m}}, \qquad |Q_\theta - \Lambda_A| \le 180\frac{\Lambda_A}{\sqrt{m}}, \qquad Q \le 16\Lambda_A. \tag{C.2}$$

C.2 Fourier inversion for count-conditioned moments

For $|t| \le 1/4$, let $z_i(t) = r_i e^{\mathrm{i}t}/(1 - r_i + r_i e^{\mathrm{i}t})$ and let $c_{i,1} = z_i - r_i$, $c_{i,2} = z_i(1 - z_i)$, $c_{i,3} = z_i(1 - z_i)(1 - 2z_i)$, and $c_{i,4} = z_i(1 - z_i)(1 - 6z_i(1 - z_i))$ be the first four cumulants of $B_i - r_i$ twisted by $e^{\mathrm{i}t(B_i - r_i)}$. Because $\sum_{i=1}^m w_i b_i = 0$, the twisted cumulant sums $M_j(t) = \sum_{i=1}^m b_i^j c_{i,j}(t)$ satisfy

$$\begin{gathered} |M_1(t)| \le 10\sqrt{mQ}\, t^2, \qquad |M_2(t) - Q_\theta| \le 20Q|t|, \qquad |M_2(t)| \le 6Q, \\ |M_3(t)| \le 20Q^{3/2}, \qquad |M_4(t)| \le 100Q^2. \end{gathered} \tag{C.3}$$

With $F_\theta(t) = \mathbb{E}_\theta e^{\mathrm{i}t(K_0 - k)}$, we have $|F_\theta(t)| \le e^{-mt^2/50}$ on $[-\pi, \pi]$ and $\mathbb{P}_\theta(K_0 = k) \ge 1/(3\sqrt{m})$ by Lemma A.1. Dividing $(2\pi)^{-1}\int_{-\infty}^{\infty} e^{-mt^2/50}|t|^j\, dt$ by $\mathbb{P}_\theta(K_0 = k)$ gives the upper bounds

$$7, \quad 25m^{-1/2}, \quad 175m^{-1}, \quad 13125m^{-2}, \quad 3 \cdot 10^8 m^{-4} \tag{C.4}$$

for $j = 0, 1, 2, 4, 8$, while on $1/4 \le |t| \le \pi$ the remaining product of at least $m - 4$ Bernoulli factors is bounded by $e^{-m/810}$. Integrating the exact Fourier integrands $F_\theta M_1$, $F_\theta(M_1^2 + M_2)$, and $F_\theta(M_1^4 + 6M_1^2M_2 + 3M_2^2 + 4M_1M_3 + M_4)$ over $[-\pi, \pi]$ yields

$$\begin{gathered} |\mathbb{E}(L \mid K_0 = k)| \le 2000\frac{\sqrt{Q}}{\sqrt{m}}, \\ |\operatorname{Var}(L \mid K_0 = k) - Q_\theta| \le 2000\frac{Q}{\sqrt{m}}, \qquad \mathbb{E}(L^4 \mid K_0 = k) \le 2000Q^2. \end{gathered} \tag{C.5}$$

Combining (C.2) and (C.5) proves (5.5)–(5.6), and comparing the one-coordinate deleted count mass to $\mathbb{P}_\theta(K_0 = k)$ via Lemma A.1 and (C.1) proves (5.7).

C.3 Proof of the one-sided clipping inequality (5.14)

Following He and Cheng [5, Lemma 3.2 and equation (83)], suppose first that $\mathbb{E}V_0 = 0$ and $|V_0| \le a := 1/10$. For $u \ge 0$, $\mathbb{E}(V_0)_+ \le u + a\mathbb{P}(V_0 > u)$ and $f(u) \ge 3u/4$ imply $\mathbb{P}(V_0 > u) + f(u) \ge \frac{3}{4}\mathbb{E}(V_0)_+ = \frac{3}{8}\mathbb{E}|V_0|$. For $u = -r$ with $0 < r \le a$, putting $t = \mathbb{P}(V_0 > -r)$, $\mathbb{E}V_0 = 0$ gives $0 \le at - r(1 - t)$, so $r \le 2at$ and $\mathbb{E}|V_0| = 2\mathbb{E}(V_0)_+ \le 2at$, whence $t + f(-r) \ge t - 5r/4 \ge 3t/4 \ge \frac{3}{8}\mathbb{E}|V_0|$. For $a < r \le 13/20$, $\mathbb{P}(V_0 > -r) = 1$ and $1 + f(-r) \ge 3/16 \ge \frac{3}{8}\mathbb{E}|V_0|$.

For general centered Y with $M_4 = \mathbb{E}Y^4 \le 1/160000$, truncate Y at $\pm 1/20$ to obtain T, and set $b = \mathbb{E}T$ and $V_0 = T - b$. By Markov's inequality, $|b| \le 8000M_4 \le 1/20$, $\mathbb{P}(T \ne Y) \le 160000M_4$, $|V_0| \le 1/10$, and $\mathbb{E}|V_0| \ge \mathbb{E}|Y| - 16000M_4$. Applying the bounded case to V_0 at $u - b \in [-13/20, 13/20]$ gives the first inequality in (5.14) with error $(3/8)16000 + 160000 + (5/4)8000 = 176000$, and applying it to $-Y$ and $-f(-\cdot)$ gives the second.

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